

Health Disparities in Nephrology

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DISPARITY

The term “health disparity” refers to a suboptimal health process or health outcome experienced by demographically defined groups that occurs in the context of social, economic, and/or environmental inequality.¹ This chapter covers four broad areas of health disparities in nephrology: 1. structural origins of health disparities; 2. health disparities in the incidence and treatment of chronic kidney disease (CKD) and its risk factors; 3. health disparities in the incidence and treatment of end-stage kidney failure (KF); and 4. initiatives to reduce health disparities in nephrology.

Global estimates of treated KF incidence vary widely and are strongly influenced by the availability of funding for the treatment of kidney diseases and associated risk factors.² Worldwide, approximately 2.6 million people received treatment for KF in 2010; however, estimates suggest that 2.3 to 7.1 million people may have died from KF due to the inability to access kidney replacement therapy.³ These numbers are projected to at least double by 2030, when 5.4 million people are expected to receive treatment for KF and 14.5 million people may die from KF due to social, economic, geographic, and other constraints.³ In many countries, members of historically marginalized groups (e.g., people of low socioeconomic status, rural/remote-dwelling residents, and members of minoritized race and ethnic groups) experience a disproportionately high incidence of KF but are less likely to receive optimal treatment for kidney failure.^{4,8} Often, structural and social determinants of health intersect to perpetuate these disparities such as in Australia, where Indigenous Australians experience high rates of poverty and geographic isolation and an approximate 10-fold higher risk of KF than nonindigenous Australians.^{7,8} Similar disparities in social conditions and in the incidence, treatment, and risk factors

of KF are observed among minoritized and marginalized groups from the United Kingdom, North America, New Zealand, and other parts of the world.⁹⁻¹⁹ The magnitude and persistence of these health disparities have prompted some governments to prioritize their elimination while also attempting to limit the overall burden and costs attributed to treating KF.^{1,20}

HEALTH JUSTICE

In 1966, Dr. Martin Luther King, Jr. proclaimed, “Of all the forms of inequality, injustice in health care is the most shocking and inhumane.”²¹ The pursuit of equal health outcomes or equal access to health care plays a prominent role in contemporary debates on health care.^{22,23} Examples of equal access and equal outcomes with respect to health care reflect the Aristotelian principles of horizontal and vertical equity.²⁴ However, such debates in health care often include conflicting views of justice. For example, a medical model of justice focuses attention at a time when someone who is already ill seeks access to scarce and/or an expensive service such as dialysis or kidney transplantation. (Many credit the birth of bioethics to the introduction of the arteriovenous shunt by Dr. Belding Scribner in 1960, which allowed for long-term hemodialysis to treat patients with KF.)²⁵ By contrast, a social model of justice highlights how social determinants such as access to education, employment, preventative care, safe neighborhoods, and nutritional foods affect the health of a population.^{23,26} In other words, the social model focuses on how a person’s need for health care arose in the first place. Although inequities in distributive justice (e.g., in kidney transplantation) persist in many areas of medicine, the nephrology community has focused growing attention on the link between structural inequities,

social conditions, and risk factors that place people at risk for progressive diseases of the kidney.^{22,27}

STRUCTURAL ORIGINS OF HEALTH DISPARITIES IN NEPHROLOGY

Structural racism refers to discriminatory policies, laws, and practices that systematically and inequitably allocate services, resources, and opportunities in society based on race (and/or ethnicity).²⁸ The resultant unequal distribution of power and resources—political, social, institutional, and economic—form the foundation of most health disparities in medicine and nephrology today.²⁹ Frequently, these inequalities have compounded over time and limit the ability of historically marginalized groups to build power and resources over generations. For example, discriminatory housing practices, forced displacement, and mass incarceration have led to marked inequities in the social environment in many parts of the world as evidenced by racially, economically, and geographically segregated communities³⁰⁻³²; differential access to employment opportunities, high-quality schools, nutritious food, and safe housing³³⁻³⁵; and marked concentration (or lack) of medical care for marginalized groups.^{26,27,36-38} Link and Phelan originally referred to social conditions, including social class, race, income, and education level as “fundamental causes of disease because these conditions govern access to resources that influence health and disease.”³⁰ Globally, race and ethnic disparities in kidney disease and its risk factors are well documented among historically minoritized groups including members of Black, Hispanic, and Indigenous groups (the terminology “Indigenous” refers to people of Aboriginal, First Nations, Native American, Native Hawaiian,

Torres Strait, and other Pacific Islander backgrounds) who, despite bearing a higher burden of KF, are less likely to receive dialysis and kidney transplantation.⁹⁻¹⁷ While these disparities arise from a variety of mechanisms including biologic susceptibility, the legacy of structural discrimination as manifested through economic vulnerability, geographic isolation, and contextual impediments to healthy living plays a central role.^{5,39-43} Moreover, the interconnected nature of social categorizations (i.e., intersectionality), such as socioeconomic status, race, ethnicity, language, religion, class, gender, and gender identity, creates overlapping and interdependent systems of discrimination that results in further disadvantage for affected individuals and groups.⁴⁴ Patients with CKD, who have backgrounds in several marginalized social categories, frequently experience multiple layers of discrimination and the worst health outcomes.⁴⁴ In the United States, for example, Black Americans experience disproportionately high rates of KF^{2,6} and face more than twice the rate of poverty of non-Hispanic White Americans.⁴⁵ Perhaps unsurprisingly, the highest incidence of KF occurs among Black Americans in the most impoverished neighborhoods, many of which were historic areas of “redlining”—a federally sponsored housing policy in the United States that permitted risk-based mortgage lending based on the racial makeup of a neighborhood.^{40,46} Of note, most studies pertaining to race and ethnic disparities in kidney disease emanate from North America, Australia, and New Zealand, where health data have historically included race and ethnicity, unlike in many European countries, where until recently health statistics were commonly organized according to occupational or class hierarchies. Discussion of the hypothesized influences of structural and social determinants of health in nephrology uses the framework shown in Fig. 81.1.^{28,47}

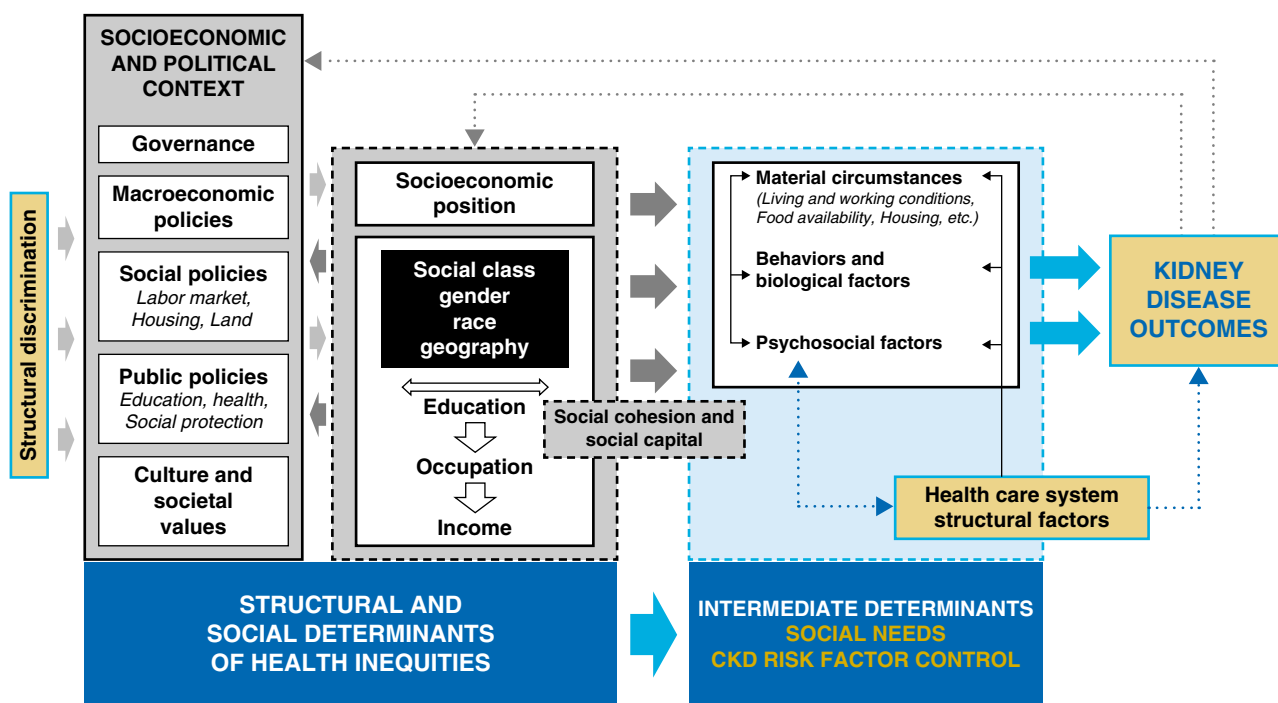


Fig. 81.1 Social determinants of health disparities framework. (From Hall YN. Social determinants of health: addressing unmet needs in nephrology. *Am J Kidney Dis.* 2018;72(4):582–591 and Solar O, Irwin A. A conceptual framework for action on the social determinants of health. WHO Report, page 6, 13 July 2010, Geneva, Switzerland 2010.)

PRENATAL ENVIRONMENT AND DISPARITIES IN RISK FACTORS FOR KIDNEY DISEASE

In 1990, Barker noted, “The womb may be more important than the home” in terms of the origins of adult disease.⁴⁸ Worldwide, every year approximately 13 to 14 million newborns (10 per 100 live births) are born prematurely and 23 million (16 per 100) weigh <2500 g (i.e., low birth weight).^{49,50} While intrauterine growth retardation and low birth weight are invariably linked with measures of maternal poverty,⁵¹⁻⁵³ much debate surrounds the influence of early fetal programming on subsequent risk of chronic diseases including those of the kidney (see Chapter 20 for a detailed discussion). The “critical periods” model is one of three models that have been proposed to explain how socioeconomic position over the life course may influence subsequent health.⁵⁴ The “fetal programming” hypothesis is based on the critical periods model and states that intrauterine growth retardation results in permanent alterations in fetal physiology that become deleterious later in adult life.⁴⁸ Brenner hypothesized that retardation of renal development as occurs in individuals of low birth weight increases postnatal risks for systemic and glomerular hypertension and predisposes an individual to further nephron loss and the development of progressive kidney injury.⁵⁵⁻⁵⁷ This hypothesis draws support from reports of 1. a direct relationship between birth weight and nephron number,⁵⁸ 2. an inverse relation between birth weight and hypertension later in life,^{48,59-61} and 3. an inverse relation between nephron number and blood pressure.⁶²

In addition to hypertension, low birth weight has been variably linked with risk factors such as diabetes and obesity and has been directly associated with later-life onset of microalbuminuria, reduced estimated glomerular filtration rate (eGFR), and KF.⁶³⁻⁶⁶ One study conducted on a cohort of term singletons born during the 1944–1945 Dutch famine found that midgestational exposure to famine was associated with microalbuminuria and that late gestational exposure was associated with glucose intolerance.⁶⁴ Notably, a rapid increase in nephron number occurs in midgestation, which is a critical period for determining nephron endowment at birth.⁶⁴ Moreover, rapid growth and weight gain during infancy and childhood that occur after intrauterine growth restriction (and/or low birth weight) are associated with increased risks of insulin resistance, obesity, diabetes, and hypertension.^{55,66,67}

Because low birth weight is strongly linked to indices of parental poverty and occurs more commonly among socially marginalized groups, researchers have posited that low birth weight may partly account for disparities in congenital, hypertensive, and hyperfiltrative kidney diseases.^{68,69} As noted earlier, Indigenous Australians experience higher rates of poverty and geographic isolation and an approximate 10-fold higher risk of KF than non-Indigenous Australians.⁷ Studies conducted in a remote Indigenous Australian community (where low-birth-weight infants compose approximately 25% of live births) observed that low as compared with high birth weight was significantly associated with lower kidney volumes, higher blood pressure, and later in life, lower eGFR and higher levels of urinary albumin excretion.^{62,70} In the Bogalusa Heart Study, a biracial prospective cohort study of cardiovascular risk

factors in children and adolescents in the southern United States, the prevalence of low birth weight was higher among Black than White infants.⁷¹ Subsequent reports have linked lower birth weight in the Bogalusa cohort with higher blood pressure levels later in life. These reports have also noted that the magnitude of the association between lower birth weight and higher blood pressure appears to increase with advancing age.^{59,60,71} The trend toward increasing survival of low-birth-weight infants further underlines the importance of intrauterine fetal environment and timing of insults in future risk for developing precursors to progressive kidney disease.⁶⁸

POSTNATAL ENVIRONMENT AND DISPARITIES IN RISK FACTORS FOR KIDNEY DISEASES

Over the past 2 decades, the nephrology community has paid increasing attention to how a person's living environment may influence downstream risk of chronic diseases. This heightened attention is evidenced by the rise in studies that examine the complex interplay of social determinants at multiple levels (individual, community, institution, society) on important processes and outcomes related to the CKD.^{40,72-74} Recognition that kidney health is intricately tied to social conditions across the life course is further reflected in the Sustainable Development Goals that were adopted by the United Nations member states in 2015.⁷⁵ People residing in underresourced neighborhoods experience more crime, less access to high-quality educational and employment opportunities, and poorer health outcomes in comparison with people from less distressed areas.^{34,73} In fact, the Black Lives Matter Movement formed to build local community power in direct response to systematic violence inflicted on, and social inequities suffered by, Black communities.⁷⁶

Several contemporary studies have described higher incidence of KF, as well as higher prevalence of albuminuria and reduced eGFR in areas of high (versus low) residential poverty.^{40,46,72,74} However, debate continues around how and how much adverse social conditions influence a person's subsequent risk of developing chronic diseases of the kidney.²⁷ Social conditions act through a host of intermediaries such as education, housing, and diet, thereby molding health-related behaviors, which in turn influence risk for kidney disease.^{27,30,73} Prior studies suggest that education may influence disparities in health behaviors through its effects on lifestyle, social support, and social norms, whereas income may drive disparities in health outcomes through access to resources and services including health care.^{77,78} The social distribution of important risk factors for progressive kidney disease including the higher prevalence of smoking, low birth weight, type 2 diabetes mellitus, hypertension, and obesity within poor as compared with less impoverished areas, lends evidence to these claims.^{22,27} This unequal distribution of intermediary exposures linked to progressive kidney injury is further compounded by the vulnerability of certain groups (e.g., persons of low socioeconomic position, rural/remote-dwelling residents, and members of minoritized race and ethnic groups) to health-compromising illnesses such as KF. Poor health then exerts potent feedback on an individual's socioeconomic position through lost income or medical costs associated with a major illness (see Fig. 81.1).⁷⁹ Recent studies have focused on increasing

screening for individual-level social risks and health-related social needs.⁸⁰⁻⁸² These studies have highlighted potential flaws in using area-level structural measures for individual-level social risks as they are commonly incongruent (e.g., not all individuals in a community that lacks safe and affordable housing experience unsafe housing or homelessness).⁸⁰⁻⁸²

The concept of “allostatic load,” the accumulation of physiologic insults due to repeated insults or chronic stressors experienced in daily life, has gained increasing interest as a model for explaining socioeconomic, geographic, and race and ethnic disparities in care processes and outcomes of chronic diseases.^{83,84} Specific measures of allostatic load include levels of hormones secreted in response to stress and biomarkers that reflect the effects of these hormones on the body.^{83,85} This concept, which is based on the cumulative exposures theory, posits that stress may accumulate from early life and manifest as an accumulation of physiologic dysregulation.^{83,86} Chronic stressors such as food insecurity, discrimination, living in substandard housing, inadequate access to health care, and exposure to violence are more pronounced among persons of lower socioeconomic status than among those of intermediate or higher socioeconomic status.^{84,85} Such psychosocial stressors may lead to hypertension and autonomic dysregulation (e.g., diminished heart rate variability), which in turn are linked to increased risk of incident KF.^{84,85,87} Accordingly, higher levels of exposure and adaptation to life stressors experienced more commonly by the poor or members of some minoritized groups may put them at higher risk for kidney disease through the aforementioned pathways than their less impoverished or referent majority counterparts.⁸⁸⁻⁹⁰

Similarly, contemporary debates in many high-income countries have increasingly focused on the role of nutrition and childhood obesity in later-life risks for development of chronic disease.^{91,92} Estimates suggest that more than a third of children in North America and Europe and approximately a quarter in the western Pacific and southeast Asia are overweight or obese and that the prevalence of childhood obesity is higher among minoritized (vs. majority) groups.⁹²⁻⁹⁴ Public health policy and nutrition experts attribute the marked growth in childhood obesity to the widespread availability of energy-dense, inexpensive foods and the low energy requirements of modern daily life.⁹³ Studies have also linked residence in impoverished areas with higher levels of processed food consumption (rich in sodium and phosphorus), lower levels of physical activity, and increased risk for chronic diseases such as cardiovascular disease, diabetes, and hypertension in comparison with residence in more affluent areas.^{35,73,95-98} In addition to offering fewer options for affordable, nutritious foods, poorer neighborhoods tend to have fewer (or less attractive) areas to promote physical activity such as parks and community centers.^{96,97} The undue influence of crime, poor housing quality, and suboptimal educational and medical infrastructure may also render poor communities less supportive of healthy lifestyles than less distressed neighborhoods.³⁴ One study linked race and ethnic disparities in childhood obesity with differential exposure to risk factors during early life, such as gestational diabetes, rapid infant weight gain, excessive television viewing, and high intake of sugar-sweetened beverages or fast food.⁹¹ The strong relation between childhood

obesity and later-life antecedents of hypertensive and hyperfiltrative kidney diseases (adult obesity, hypertension, metabolic syndrome, type 2 diabetes mellitus) suggests that interventions that target the early life period may contribute substantially to reducing socioeconomic, geographic, and race and ethnic disparities in kidney and other chronic diseases.^{70,98,99}

Perhaps the strongest evidence that neighborhood resources influence downstream risk factors and lead to health status disparities in kidney disease emanates from the Moving to Opportunity Study.^{31,33} During 1994–1998, the U.S. Department of Housing and Urban Development conducted a “randomized social experiment” in five cities (Baltimore, Boston, Chicago, Los Angeles, and New York City) to examine the extent to which relocating from a more to a less distressed neighborhood might influence downstream health outcomes. Collectively, 4498 women (90% were either Black or Hispanic women) with children from high-poverty neighborhoods were randomly assigned to one of three groups: 1788 were assigned to receive housing vouchers that were redeemable only if they relocated to a low-poverty area, 1312 received unrestricted vouchers, and 1398 were assigned to a control group that offered neither of these opportunities. In terms of risk factors for kidney diseases, the study found that participants who received a voucher to relocate to a low-poverty neighborhood experienced absolute risk reductions of 5% and 4% for extreme obesity and diabetes mellitus, respectively, 10 to 15 years later in comparison with controls.³¹ Notably, the effects of relocation occurred without detectable effects on individual-level income.¹⁰⁰ However, the strong correlation between neighborhood characteristics (e.g., crime, educational and employment opportunities, recreation, and social norms) makes the evaluation of specific pathways to these outcomes elusive.²⁶ Although provocative, the results of the Moving to Opportunity Study highlight the challenges faced by policy makers and clinicians in addressing the legacy of structural racism and more proximal mediators of disparities in important CKD risk factors.

RACE, SOCIAL CONDITIONS, AND DISPARITIES IN INCIDENT KIDNEY DISEASE

Regrettably, in many parts of the world, structural discrimination and systemic bias have resulted in inequitable living environments frequently based on race or ethnicity. Link and Phelan³⁰ emphasize that race is itself so closely tied to social and economic resources that it should also be considered a fundamental cause of health inequality. Claims that social disadvantage largely explains disparities in risk for progressive kidney diseases among marginalized groups, for example, are supported by several studies.¹⁰¹ Among 603 children (median age 10 years) with nonglomerular kidney diseases enrolled in the CKD in Children study, the significantly shorter time to KF onset among Black as compared with non-Black children (median difference 3.2 years) was attributable to indices of socioeconomic status.¹⁰² In adults, nearly three-quarters of the excess risk of incident albuminuria or reduced eGFR among Black in comparison with White participants in the Atherosclerosis Risk in Communities Study was likewise attributable to measures of socioeconomic status and health care access.^{103,104}

However, studies in other health care settings indicate that a substantial fraction of the excess risk of incident KF among Black and White Americans remains unexplained even after one has accounted for differences in the distribution of KF risk factors, measures of socioeconomic status, and access to health care.¹⁰⁵⁻¹⁰⁷ The persistence of marked racial disparities in KF risk within settings where socioeconomic gradients are theoretically reduced, along with contemporary genetic data (see later) linking West African ancestry to certain forms of progressive kidney disease, further highlights the complex influences of gene–environment interactions on these disparities.¹⁰⁷⁻¹⁰⁹

Our understanding of disparities in kidney diseases among other minoritized race and ethnic groups is less developed, but the origins of these disparities share common social constructs. In North America, for example, the incidence of KF is substantially higher among Indigenous Americans (primarily among American Indians and Indigenous Canadians) than among non-Indigenous Americans.^{2,16,17,110} This situation is mirrored in the Indigenous people of Australia and New Zealand who frequently reside in rural or remote communities.⁷ Differences in the distribution of major CKD risk factors, such as obesity, diabetes mellitus, and hypertension, account for most of this excess risk of KF among indigenous people.^{7,13,110} In fact, long-term cohort studies of Indigenous Australians suggest that the lifetime risk of KF among people with diabetes may be as high as 41% in this group.¹¹ Moreover, the remote location of many indigenous communities combined with cultural differences in their understanding of illness often hinders the implementation of interventions to adequately address CKD risk factors.^{10,13,15,111,112} Notably, the U.S. Indian Health Services began systematic efforts in the 1990s to identify, screen for, and address risk factors—primarily blood pressure and diabetes control—among American Indians (Fig. 81.2).¹⁴⁻¹⁶ These efforts have been widely touted as the drivers of the decline in the incidence of diabetes-related KF

among American Indians over the ensuing 3 decades (Fig. 81.3).^{2,14,113}

BIOLOGICAL ORIGINS OF DISPARITIES IN NEPHROLOGY

Although differences in the built environment undoubtedly play a central role in promoting race and ethnic disparities in the incidence and treatment of kidney disease, epidemiologic and genetic studies support the presence of strong biologic influences as well.^{109,114} Notably, the higher KF incidence experienced by members of minoritized race and ethnic groups in comparison with White referent groups exists even within universal health care systems (with theoretically equal care access), as well as within underserved health settings (in which socioeconomic gradients are theoretically attenuated).^{107,108,115} The persistence of race and ethnic differences in KF risk in these studies suggests that factors beyond socioeconomic status, geography, and health care access contrive to influence kidney disease progression than previously inferred.^{109,114}

Certain types of kidney disease are hypothesized to occur among susceptible individuals through a complex interplay of genetic predisposition and environmental exposures.^{109,114,116} Shared exposures from the familial environment such as poverty, nutrition, and health-related behaviors may contribute to familial risks for kidney disease, but further studies have linked *APOL1* gene mutations with both acute kidney injury and chronic kidney diseases such as focal segmental glomerulosclerosis and human immunodeficiency virus (HIV) nephropathy.^{109,117,118} Estimates suggest that as much as 40% of the increased burden of progressive nondiabetic glomerular diseases in sub-Saharan African ancestry populations might be eliminated if the injury mechanism for *APOL1* risk variants could be overcome.¹¹⁹⁻¹²¹ Please refer to Chapters 18 and

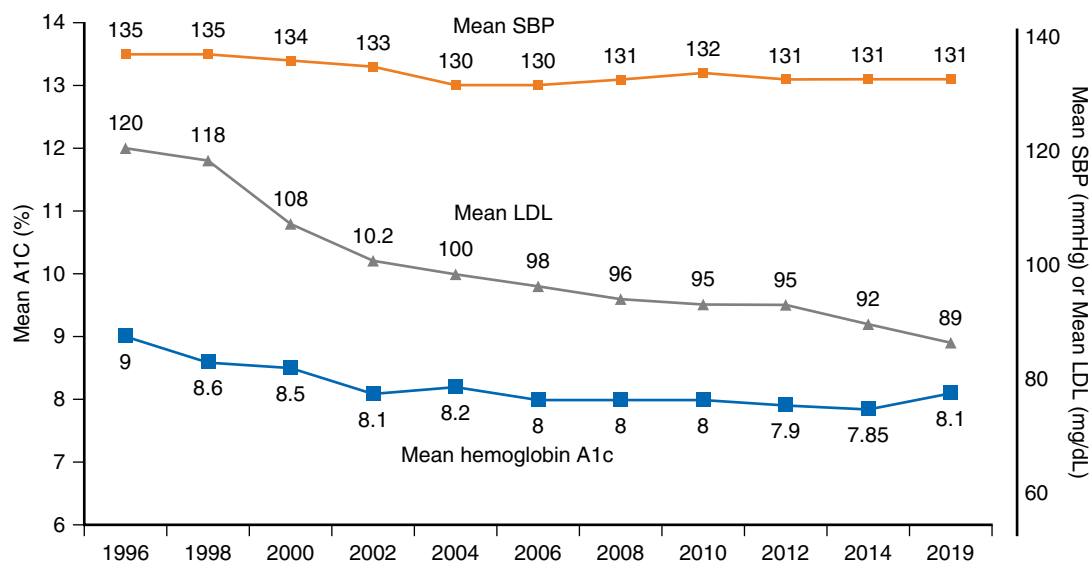


Fig. 81.2 Risk factor control among American Indians with diabetes mellitus within the U.S. Indian Health Service. (From Bullock A, Burrows NR, Narva AS, et al. Vital signs: decrease in incidence of diabetes-related end-stage renal disease among American Indians/Alaska Natives—United States, 1996–2013. *MMWR Morb Mortal Wkly Rep.* 2017;66(1):26–32.)

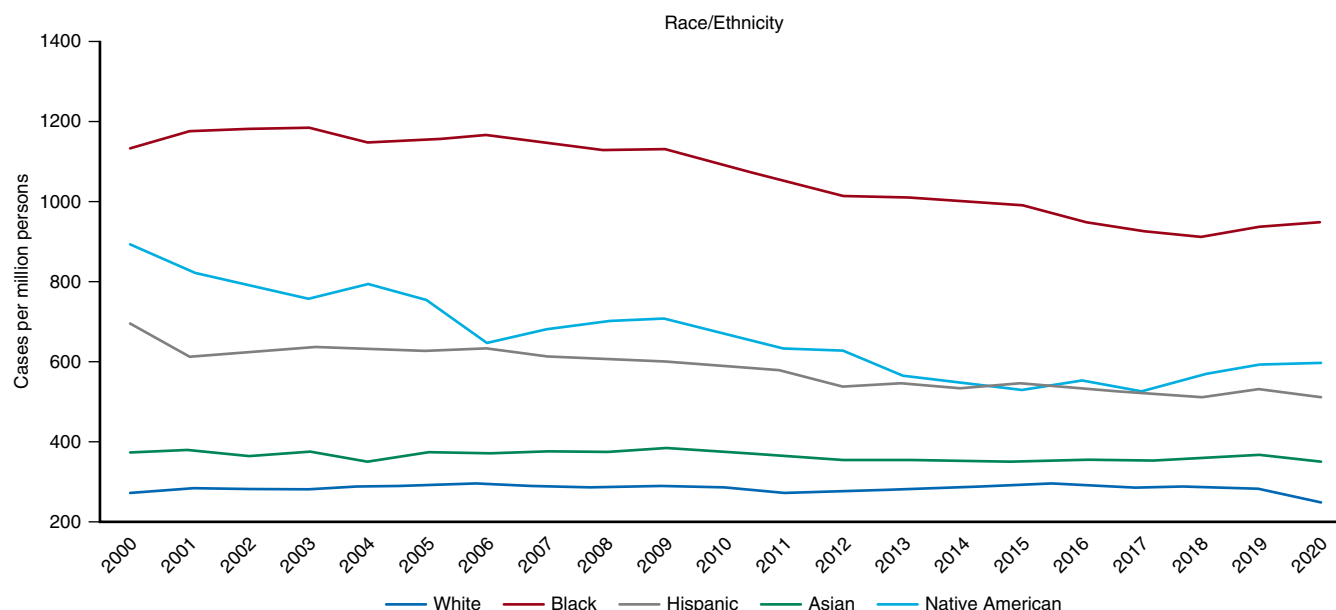


Fig. 81.3 Trends in age- and sex-adjusted incidence rate (per million/year) of end-stage kidney failure by race and ethnicity in the United States, 2000–2020. (From Johansen KL, Chertow GM, Gilbertson DT, et al. U.S. renal data system 2022 annual data report: epidemiology of kidney disease in the United States. *Am J Kidney Dis.* 2023;81(3S1):A8–A11.)

44 for further discussion of the complex relations among ancestry, genes, and related diseases of the kidney.

HEALTH SYSTEM AND DISPARITIES IN NEPHROLOGY

More than 40 years ago, Donabedian proposed a model for assessing the quality of medical care based on structure, process, and outcome. According to the Donabedian model, “process—the method by which health care is provided—is limited by the structure or environment in which it operates.”¹²² The health system plays an important role in mediating disparities in nephrology. Health systems are responsible for promoting and implementing interventions to improve health by altering exposure to key intermediaries. However, in many countries, differential access to the health system and differential access to high-quality, integrated health care often lead to relative differences in the receipt of key interventions to promote and maintain health (see further discussion in Section 13, Global Considerations in Kidney Disease).

UNEQUAL HEALTH SYSTEM ACCESS AND SURVEILLANCE

In 1971, Julian Tudor Hart observed in his landmark paper “The Inverse Care Law” that disadvantaged populations needed more health care than advantaged populations but received far less.¹²³ Disparities in the surveillance and treatment of CKD risk factors likely contribute to the higher incidence of more advanced CKD in marginalized populations. Persons with CKD from underresourced communities are often difficult to identify because few countries (including those with high per capita health care spending) track the care of poor or uninsured

patients.^{2,4,113,124} In many instances, CKD can be slowed or prevented. Interventions such as blood pressure lowering, use of inhibitors of the renin–angiotensin system and sodium–glucose transporter-2, and avoidance of nephrotoxins are effective in reducing morbidity and slowing progression of most kidney diseases.^{125–131} However, differences in health system access (e.g., according to socioeconomic position) limit the time and resources needed to effectively implement these interventions in the target populations, as evidenced by underutilization of these therapies among marginalized groups.^{132–134}

In countries without universal health care, such as the United States, the scope of basic health insurance coverage plays a critical role in the treatment of modifiable risk factors for kidney diseases.¹³² The presence and scope of basic health insurance is also linked to pre-KF nephrology care, mortality, and preparedness for KF (i.e., rates of home dialysis usage, kidney transplantation, and hospitalization and venous catheter usage at dialysis initiation).^{135,136} For example, Kurella-Tamura and colleagues¹³⁷ observed a significantly lower incidence of KF and smaller health insurance–related gaps in pre-KF nephrology care access among states with broader as compared with narrower health insurance coverage for the poor. In 2014, USA policymakers and health care providers heralded expansion of basic health insurance coverage for low-income people in the United States under the Affordable Care Act.¹³⁸ In terms of pre-KF nephrology care access, states that expanded coverage under the Affordable Care Act witnessed 59% more preemptive listings for kidney transplantation among low-income people than states that did not expand coverage. There were also proportionately larger increases in basic health insurance coverage among preemptively listed people of minoritized race and ethnic groups as compared with those of non-Hispanic White race.¹³⁹

As noted earlier, prior studies showed that lack of health insurance coverage, residential poverty, and non-White race or ethnicity were independently associated with worse metabolic abnormalities at dialysis initiation, including more severe hypoalbuminemia, azotemia, and anemia, and with marked delays in accessing the transplant waiting list and in receiving a kidney transplant.^{32,42,43} The previously mentioned reports suggest that marginalized groups receive less and perhaps substandard care in earlier stages of CKD.^{73,140} Although differences in individual-level factors such as adherence to screening and treatment recommendations may partly contribute to these disparities, it is likely that differential access to the health system plays a central role in determining which groups receive these interventions, particularly at earlier stages of CKD.¹⁴⁰ However, as Schrag¹⁴¹ noted in 1983, the ability to pay for health services does not necessarily guarantee their availability. Differential access to health services is also well documented within high-income countries with universal health care, such as in the United Kingdom, where the age- and sex-adjusted prevalence of severe CKD (eGFR <30 mL/min per 1.73 m² at presentation) was 6- to 9-fold higher in areas of Sheffield with the highest (vs. lowest) deprivation. The higher risk of severe CKD in impoverished areas was largely attributable to differences in the distribution of diabetes and hypertension.^{5,142}

IMPORTANCE OF HEALTH SYSTEM STRUCTURE

The medical care of most marginalized groups is concentrated in the hands of relatively few providers.³⁶ For example, owing to their relatively limited options for ongoing ambulatory care, many marginalized people (low-income, underinsured, members of minoritized race and ethnic groups, refugees, homeless, and those with specialized care needs such as addiction medicine and mental health) in the United States seek ambulatory care from public hospitals and safety net health care clinics.^{143,144} Even within universal health care systems such as in Canada or Australia, the rural or remote nature of some minoritized communities strongly influences the availability, frequency, and quality of care received.^{10,11,13} The centralization of services that disadvantage rural and remote patients is especially apparent in transplantation, where most specialists and transplant centers are located in large urban centers.¹⁴⁵ Several studies demonstrated that socioeconomic, geographic, and race and ethnic disparities in care may be partly explained by differences in where and by whom patients are treated, leading to systematic differences in the care environment, provider quality, and access to care.^{36,38,146} For example, Bach and associates³⁶ found that Black and White Medicare beneficiaries were treated for the most part by different primary care physicians. Moreover, the primary care physicians who cared primarily for Black patients were less likely to be board certified, more likely to be in training, and more likely to report encountering system barriers to providing high-quality care to their patients (e.g., securing referrals to specialists). Varkey and colleagues³⁸ further found that clinics serving higher proportions of patients from minoritized race and ethnic groups were more chaotic than, and had different organizational characteristics from, those of private clinics.

Recent qualitative studies of nephrology and primary care providers have identified system-level factors such as referral process delays, insufficient time to build patient partnerships, and poorly integrated nephrology and primary care as major barriers to optimal preparation for KF in diverse health care settings.^{147,148} Such systematic barriers may also account for the historically lower utilization of home dialysis and preemptive kidney transplantation among Black, Hispanic, and rural-dwelling patients than among non-Hispanic White and urban-dwelling patients, respectively.^{149,150} In short, several key factors linked to the provision of high-quality medical care, such as coordination of care, subspecialist access, and provider experience differ markedly according to the health system. Such health system-level determinants may delay or restrict the access of marginalized patients to receiving adequate prevention and management of kidney disease and its antecedent risk factors.

IMPACTS OF GEOGRAPHIC LOCATION

People from rural/remote communities commence their journey with CKD from a disadvantaged position because they tend to have fewer economic resources, less formal education, and lower access to primary and specialist health care than people from urban communities.¹³⁴ They face many barriers to access kidney replacement therapies, including late referrals to specialist nephrologist care, forced relocation to access treatment, transportation challenges, and financial hardship (Table 81.1). These factors often intersect and contribute to an increased risk of mortality, morbidity, and hospitalization compared with their urban counterparts.^{145,151} For patients on maintenance dialysis, longer travel time to dialysis (>50 km) is associated with higher mortality, higher and earlier risk of peritonitis for people on peritoneal dialysis, and decreased quality of life.¹⁵² Rural and remote-dwelling patients encounter far more barriers to accessing kidney transplantation due to the highly centralized nature of specialist care and transplant centers, which are universally located in large cities. Logistic barriers range from costly transportation and housing to the cultural impacts of relocation and isolation.¹⁵¹ Among patients with advanced kidney disease, rates of relocation (for care) increase with higher degrees of residential remoteness: Low prevalence of in-center hemodialysis facilities in remote areas is a primary driver of this medical relocation.¹⁵¹ The adverse effects of permanent or medical relocation for rural and remote-dwelling patients and their families can be particularly challenging for Indigenous People of Canada and Australia (see Table 81.1).^{153,154} These effects include cultural isolation, feeling homesick, separation from family and friends, and psychosocial stress, which can further impact health outcomes.

DISPARITIES IN THE TREATMENT OF CHRONIC KIDNEY DISEASE

Blood pressure control is a central target to reduce morbidity and mortality in CKD.¹²⁸ Hypertension is the attributed cause of more than a quarter of incident cases in the United States.^{2,113} Hypertension is also the most common modifiable risk factor for disease progression among uninsured

Table 81.1 The “Illness Experience” of Patients with End-Stage Kidney Disease from Qualitative Studies in Diverse Communities

Social Category	Country	Context/theme	Quotation
Rural-dwelling patient (Indigenous)	Australia	Maintenance dialysis Geographic relocation	“My community is 1000 km from Darwin. I’ve been on dialysis now for 3 years, and I am based here (Darwin) now.” ²³⁰
Rural-dwelling patient	Australia	Maintenance dialysis Geographic relocation Lack of safe and sustainable resources	“We’ve got a patient ... that we trained on peritoneal dialysis, but he’s currently living in Brisbane with his daughter because we cannot get supplies to him.” (nephrologist) ²³¹ “Water quality ... We have had a patient who wanted to do home hemo[dialysis] and we were breaking [reverse osmosis systems] trying to purify the water to the point where she couldn’t do it.” (nurse) ²³¹
Rural-dwelling patient	New Zealand	Maintenance dialysis Geographic isolation	“If I could do the dialysis here (at home) I would choose to have it, it’s not that I actually want to die ... I can’t do it myself, so it’s travel, move or die, and dying seems like the best of a bad bunch for me.” ²³²
Remote-dwelling patient (Indigenous)	Australia	Geographic isolation and cultural shock	“It’s three or four planes from Thursday Island to get down to Brisbane or helicopter all and hoping to get down to Brisbane. ... Some of these patients have never been on a plane before, so there’s so many other obstacles that they’re facing on top of their health.” (nephrologist) ²¹¹
Remote-dwelling patient (Indigenous)	Australia	Transplant evaluation Cultural shock	“So, when they’re going through their assessments, they got to get in personal. Put up their shirts, showing that, you may have the tummy in whatever played and poked and prodded, and they had their weight questioned. That is personal, and it can be a bit embarrassing for them to talk about and feel like they’re being told off or scolded for the way that they’ve been thinking or where their weights at, or their health situation.” (patient) ²¹¹
Immigrant Minoritized race	USA	Emergency dialysis Psychosocial and physical distress	“I would take the bus and had to struggle a lot, and they sent me back. They said that my potassium levels weren’t high enough ... they told me to come back tomorrow, and sometimes I would hold on for 2 days so that they wouldn’t send me back again (without emergency dialysis).” ¹⁵⁸ “It’s intensive care. Many of my friends that I met there (at the hospital) have already died, and they died there (at the hospital receiving emergency dialysis). ... and every time I go (to the hospital), I feel bad.”
Minoritized race	USA	Maintenance dialysis Mistrust	“My doctor(s) ... be shocked when I asked them, well why is there so many Black people on dialysis and they don’t have no real answer for me. I really don’t like that. And so then on top of that, he only spends 90 seconds with me ... I’m like wow, I feel like cattle.” ²³³
Remote-dwelling transplant recipient	Australia	Transplant complications Financial costs (in universal health insurance system)	“I know that I’ve got one complicated patient who’s had a whole lot of surgical complications post transplant. And he told me the other day, and it actually made me gasp, that him and his wife, both on the Pension, are AUS\$20,000 in the red over the last two and a half years.” ^{211,231}

American adults with CKD, affecting approximately 57% of this population as compared with obesity (40%), diabetes (22%), and overt albuminuria (13%).¹³² However, several studies have described marked disparities in blood pressure control among people with CKD according to measures of socioeconomic status and health system access. For example, data from the U.S. National Health and Nutrition Examination Surveys indicated that hypertensive adults with reduced eGFR (15–59 mL/min/1.73 m²) and/or

albuminuria (≥ 30 mg/g) who lacked health insurance were far less likely to be receiving treatment for their hypertension than insured counterparts.¹³² Moreover, among hypertensive adults with reduced eGFR and/or albuminuria, those who lacked health insurance coverage were 55% less likely to be receiving treatment with a renin-angiotensin system inhibitor than insured adults.¹³²

In addition to experiencing reduced access to high-quality care, historically marginalized groups may harbor

risk factors that heighten their susceptibility to further morbidity and mortality. Among ambulatory patients with moderate or advanced CKD (eGFR <60 mL/min/1.73 m²) from a large public health system in the United States, social risks such as chronic viral diseases, substance abuse, homelessness, and mental illness were far more prevalent than observed in other health care settings.^{108,155} In addition, approximately 7% of persons with moderate-to-advanced CKD progressed to KF and 16% died during median follow-up of 6.6 years.¹⁰⁸ Notably, nearly 30% of patients were never assessed for proteinuria, and less than 20% were ever seen by nephrology specialists during follow-up.¹⁰⁸ The elusive nature of this population, their frequent loss to follow-up, reduced access to specialty care, and limited resources to support disease surveillance further complicate effective CKD management. In summary, marginalized patients with nondialysis-dependent CKD are “invisible” to health care policymakers and health systems until late in the course of their CKD. Although evidence is scarce, in most countries, diabetes and hypertension appear to drive disparate rates of CKD progression among marginalized as compared with majority of patients.^{2,156}

DISPARITIES IN THE TREATMENT OF END-STAGE KIDNEY DISEASE

Conservative estimates suggest that KF affects at least 4.9 million people worldwide at considerable cost to society.³ In some high-income countries with universal access to KF treatment, socioeconomic, geographic, race, and ethnic disparities exist in access to certain types of kidney replacement therapies.^{2,3,113} In most medium- and low-income nations, access to KF treatment remains limited and is governed predominantly by socioeconomic position and occupation.^{3,4,157}

In terms of access to maintenance dialysis, socioeconomic, geographic, and race and ethnic disparities are evident at several levels, from a country's health region and health care system to the individual patient. Regular access to maintenance dialysis remains, for the most part, limited to patients receiving care in high-income and upper-middle-income nations.^{3,4,113} In most low-income and lower-middle-income countries, the majority of patients who initiate dialysis either die or stop treatment within the first 3 months because of cost restraints.^{4,124} While the numbers who do not receive treatment for KF is unknown, estimates suggest that at least 2.2 million people died prematurely in 2010 because kidney replacement therapy could not be accessed, with the largest treatment gaps occurring in low-income countries in Asia and Africa.³ Even within high-income nations, disparities in access to maintenance dialysis exist, such as among undocumented immigrants in the United States, who in many states can only receive emergency (versus maintenance) dialysis.¹⁵⁸

As noted previously, there are prominent and persistent race and ethnic disparities in process measures at dialysis initiation that largely reflect the presence and quality of care leading up to KF.^{42,159,160} However, data from the U.S. Centers for Medicare and Medicaid Services (CMS) suggest that these disparities attenuate and may even reverse for members of minoritized race and ethnic groups within the

first year of initiating maintenance dialysis.^{161,162} In other words, the establishment of a regular site of care with access to wraparound services (i.e., social work, dietician, and primary nephrology care) may partially offset the lack of, or variation in, medical care received before KF onset. Whether achievement of specified targets for these measures actually reflects dialysis-related quality of care remains a source of substantial debate.^{150,163} Health disparities in home dialysis usage among marginalized groups appear similarly rooted in social factors including the quality of housing and availability of basic services. For example, many urban patients from marginalized groups are unable to perform home dialysis due to unsafe living conditions, overcrowded housing, and/or inadequate space to store supplies.¹⁴⁹ For many rural/remote-dwelling patients, home dialysis is unavailable due to supply chain limitations (e.g., limited water supply, improperly treated water, or inability to deliver dialysis treatment supplies).¹⁶⁴

KIDNEY TRANSPLANTATION

In terms of survival and quality of life, kidney transplantation remains the optimal treatment for most patients who progress to KF. However, significant disparities in receiving a transplant according to socioeconomic status, geography, race, and ethnicity persist worldwide.^{9,113,136,165-168} In Canada, transplantation rates among Aboriginal, African, Indo Asian, and East Asian Canadians range one-half to two-thirds those of White Canadians despite the presence of uniform health insurance coverage.^{113,166} In Australia, transplantation rates in Indigenous Australians are 77% lower than those in nonindigenous Australians, and in New Zealand, Maori and Torres Strait Islander patients with KF are similarly disadvantaged.⁷ Similar disparities are also present in the United States, where kidney transplantation rates are significantly lower among the poor and among patients of most minoritized race and ethnic groups than among less disadvantaged groups and non-Hispanic White patients, respectively.^{22,168}

Although key mediators of, and their relative contributions to, health disparities in transplant access vary by population and health system, many are directly linked to structural discrimination (e.g., forced displacement) and the social conditions described previously.^{46,169} For example, in the United States, geographic differences account for most of the disparity in receiving a deceased-donor kidney from the waiting list among Hispanic, Native Hawaiian, Pacific Islander, American Indian, and Alaska Native patients.⁴³ By contrast, contextual poverty and health insurance coverage account for the largest fraction of the disparity in live-donor kidney transplantations among American Indian, Alaska Native, and Black and Hispanic patients in comparison with non-Hispanic White patients.¹⁶⁷ Additional studies have identified differences in attitudes and knowledge about kidney transplantation among Black and Hispanic Americans in comparison with non-Hispanic White Americans.¹⁷⁰ In Australia, only 1.9% of Indigenous patients receiving maintenance dialysis were actively listed as compared with 9.5% of non-Indigenous dialysis patients.^{171,172} A study of kidney specialists reported that concerns about poorer outcomes for Indigenous Australian patients compared with non-Indigenous patients was a major reason for nonreferral

(for transplant waitlisting).¹⁷³ The influence of geographic, cultural, and/or linguistic isolation on delays in accessing the transplant waiting list or receiving a kidney transplant remains understudied.⁴³

Marginalized patients appear to encounter more difficulty in navigating the complex steps required to successfully receive a kidney transplant, and this vulnerability appears to have changed little over time.¹⁷⁴⁻¹⁷⁶ This may be a direct effect of low health literacy, inadequate insurance coverage, suboptimal patient-provider communication, and lack of support and information about transplant, all of which have been linked to delays in accessing the transplant waiting list, in completing the transplant evaluation, and in successful receipt of a transplant.^{111,166,174,177-180} As noted earlier, kidney providers also appear reticent to refer marginalized patients for transplant because of concerns about poor outcomes or inadequate coverage for prescription and procedural costs.^{173,178,181} In a national survey of transplant programs in the United States, >70% responded that they frequently or occasionally exclude patients from the kidney transplant waitlist owing to concerns that the patients will not be able to afford the immunosuppressive medications.¹⁸²

Although socioeconomic status and contextual poverty are linked with disparities in transplantation rates in the United States,^{39,91,174} the magnitude of such disparities is markedly higher in middle-income nations such as in Mexico, where transplantation rates are more than 10-fold higher among insured patients than among uninsured patients.¹⁸³ As concerning is the unethical practice of organ trafficking and exploitation of disadvantaged populations for profit that occurs in many low- and middle-income nations.¹⁸³ This pervasive commercialization of live-donor transplantation is supported by reports of affluent foreigners undergoing large fractions of the kidney transplantations performed in some low- and middle-income countries.¹⁸⁴ Although some writers justify paying persons to donate a kidney as mutually benefiting recipient and donor, one study conducted in India showed that such transactions do not lead to long-term economic benefit and may be associated with a decline in health for the donor.¹⁸⁵

Socioeconomic, geographic, and race and ethnic disparities in kidney transplantation extend to outcomes after receipt of a functional allograft. Similar to transplant access, specific causes of these disparities in allograft survival may differ by population and health system, but the root causes commonly reflect differential social conditions and access to high-quality health care. For example, adherence to posttransplantation immunosuppressive therapy remains a critical determinant of long-term allograft survival. Regrettably, inability to afford immunosuppressive medications appears to be a primary mechanism that restricts low-income patients from not only receiving a kidney transplant but also maintaining one.^{186,187} In one study, >40% of transplant recipients reported financial difficulty after transplantation, with health-related out-of-pocket expenses averaging approximately US\$475 per month.¹⁸⁷ In a recent single-center study of 1171 kidney transplant recipients, lower socioeconomic status was associated with an 11% higher risk of acute rejection and 77% higher risk of graft loss among Black versus non-Hispanic White Americans.¹⁸⁸ Socioeconomic disparities in maintaining

a transplant are magnified in medium-income countries such as Mexico and Pakistan, where excellent long-term results for allograft survival are offset by the prohibitive cost of immunosuppressive and antiviral medications. In these countries, transplant access and allograft maintenance for the average citizen often require a state-sponsored model.^{4,60,183}

Racial differences in biologic factors such as age at transplantation, HLA mismatching, and APOL1 genotype status may account to some degree for the remaining disparity in allograft survival between some race and ethnic groups, such as between Black and White Americans.^{168,189,190} The transplant community is now seeking ways to integrate the APOL1 genotype into clinical practice to enhance evaluation of donor organs and better align this assessment with recipient needs.¹⁹¹ Although prevailing beliefs attribute reduced graft survival among Black recipients to heightened immune responsiveness and poorer immunologic matching, few reports have actually linked better HLA matching with improved graft survival in this race group.^{189,192,193} Moreover, Asian recipients enjoy the longest graft survival of all race and ethnic groups despite the presence of suboptimal HLA matching comparable with that observed in Black recipients.^{189,194} Irrespective, the influence of biologic perpetrators of race and ethnic disparities in kidney transplantation access, such as the distribution of ABO blood type and HLA, has progressively declined in the United States due to advances in the deceased-donor allocation policy.^{43,192,195}

Lastly, studies have raised concerns regarding inequities in the quality of transplanted organs based on recipient socioeconomic status and race and ethnicity.¹⁹⁶ Data on provision of comparatively “lower-quality organs” to recipients from marginalized groups than to White, more affluent, or college-educated counterparts have cast additional light on the complex nuances that underlie this social gap in allograft access and survival.^{196,197} In terms of the disparity in allograft loss between Black and White Americans, the gap has steadily declined over the past 2 decades. Recent data from the U.S. organ transplant registry point to differential improvements in 5-year graft survival after deceased-donor kidney transplantation among Black (from 49% to 69%) as compared with White (from 63% to 75%) Americans. Similar gains (63% to 79% among Black vs. 79% to 86% among White patients) have led to a narrowing of the Black-White gap in allograft survival following living-donor kidney transplantation in the United States.¹⁹⁸ Temporal changes in deceased-donor allocation, immunosuppressive agents, and desensitization strategies have been credited as major contributors to these observations.

MOVING TOWARD MORE EQUITABLE CARE

Strategies to reduce disparities in chronic diseases include three general approaches: 1. targeted programs, 2. gap programs, and 3. gradient programs.^{79,199} “Targeted programs” comprise strategies aimed at improving the health of groups that are particularly disadvantaged in terms of disease burden or treatment access. Such targeted programs

are advantageous in that they focus on a well-defined, often small segment of the population. This aspect provides for ease in monitoring and assessing results. However, the targeted approach is similarly weakened by its narrow focus and its lack of commitment to reduce the disparity between the most and least disadvantaged groups. In some instances, health gains in the targeted groups may still lag behind those observed in less disadvantaged groups, leading to widening of the disparity.⁷⁹ An example is a program that attempts to promote live-donor kidney transplantation among remote-dwelling Indigenous patients. Although such programs may increase live-donor kidney transplantation rates among Indigenous people, the disparity between Indigenous and nonindigenous patients may actually increase owing to the differential effects of other programs (e.g., paired donation) or urban resource-mediated advances (e.g., Internet use, social networking, and multicenter listing).

Gap strategies—strategies that target a health gap between two or more socially defined groups—are more challenging than those that seek to improve the health status of targeted groups. To succeed, “gap programs” must achieve absolute improvements in health status among the intervention group (e.g., persons in the lowest socioeconomic position) at a rate of improvement that exceeds that observed in the comparison group. Like targeted programs, gap programs often focus solely on the most disadvantaged groups and largely ignore gradients in health and the health status of those in the intermediary groups.⁷⁹ Gap programs, while limited in scope, benefit from lower costs and quicker start-up.

In contrast to targeted and gap programs, “gradient strategies” address the effect of inequality on health across entire social and economic hierarchies. In other words, a gradient strategy addresses not only intermediaries of health disparities among the disadvantaged but also systematic inequities in structural determinants (e.g., policies) and/or life course exposures (e.g., education, occupational opportunities, nutrition, living standards, and/or health care access) that strongly influence an individual’s position in the socioeconomic hierarchy.⁷⁹ Needless to say, gradient strategies must often contend with conflicting political agendas, major logistic challenges, and prohibitive costs.³⁷ Moreover, such strategies commonly yield results only over a prolonged time frame. These three strategies (targeted, gap, and gradient) for reducing disparities are intended to complement one another and can often provide sequential layers for addressing structural inequities.⁷⁹

On the basis of the framework shown in Fig. 81.1, researchers have identified four major intervention areas for addressing disparities in kidney disease.⁷⁹ First, reducing inequalities in the distribution of structural determinants such as education, occupation, and income may alter social stratification and mitigate its effects on health outcomes. Second, addressing specific intermediary determinants that mediate the effects of socioeconomic position on health (e.g., dietary intake, physical exercise, unhealthy living conditions, and cigarette smoking) may reduce specific health-damaging exposures that disproportionately affect disadvantaged groups; in other words, changing the distribution of risk factors related to socioeconomic position may attenuate long-term health consequences for marginalized groups.²⁷ Third, reversing the effect of health status on socioeconomic

position may lessen the vulnerability of marginalized groups to the health-damaging conditions they face²²; examples include programs to keep persons with chronic diseases such as diabetes mellitus and KF within the workforce.²⁰⁰ Lastly, targeting the delivery of health care to marginalized groups may reduce the unequal consequences of illness and prevent further disease progression and economic loss; such strategies might offer additional care or health coverage to members of marginalized groups who have conditions (obesity, diabetes, hypertension, homelessness) that are linked to the excess burden of progressive CKD in these groups.⁷⁹

INITIATIVES TO REDUCE DISPARITIES

Successful initiatives to reduce socioeconomic, geographic, and race and ethnic disparities in CKD risk factors might reference the Special Diabetes Program for Indians administered by the U.S. Indian Health Services. In the 1990s, in response to rising concerns over the burden of KF due to diabetes in many tribal communities, the Indian Health Service established the Standards of Care for Diabetes Program (followed later by the Kidney Disease Program) in order to improve the screening and management of patients with diabetes and CKD.¹⁴ Based on the Chronic Care Model, these two programs implemented routine reporting of eGFR, annual monitoring of protein excretion, utilization of renin–angiotensin system antagonists, aggressive control of glucose and blood pressure, and enhanced patient and provider education (see Fig. 81.2).^{14,201} By 2001, age-adjusted KF incidence among American Indians with diabetes had decreased by 31% from that in 1990. By 2006, 82% of American Indians with hypertension and diabetes were receiving a renin–angiotensin system inhibitor. In 2016, age-adjusted incidence rates of diabetes-related KF had decreased by 53%, from 667 per million population in 2000 to 312 per million population in 2016.^{2,14,113,202,203} A similar gap initiative by the Australian government—the Medical Outreach–Indigenous Chronic Disease Program—aims to close the disparity in life expectancy between Indigenous and non-Indigenous Australians by improving access of indigenous people to best-practice chronic disease management.¹²⁴ Notably, diabetes mellitus and CKD rank among the five areas of highest priority. The recently established Caring for Australians and New Zealanders with Kidney Impairment (CARI) guidelines notably include rules for culturally safe kidney care in Indigenous Australians and New Zealanders.²⁰⁴

In terms of targeted programs for dialysis care, the Purple House is fostering innovation in dialysis services for remote-dwelling Indigenous Australians through mobile trucks that provide essential hemodialysis services to remote communities in Central and Northern Australia.²⁰⁵ This service provides culturally safe, nurse-assisted dialysis to reduce the need for Indigenous Australians to relocate permanently or travel large distances to access dialysis.^{205,206} Services are partially government subsidized following advocacy from remote-dwelling Indigenous leaders; additional funding has been obtained through philanthropic support and other Australian government grants. As noted earlier, home dialysis uptake is frequently lower in marginalized

groups due to barriers such as overcrowded housing and lack of storage space.^{149,207,208} In New Zealand, community dialysis houses have overcome some of these disparate barriers to home dialysis by providing accessible spaces to perform hemodialysis for patients who are fully trained and independent in their dialysis care.²⁰⁷ A community-based house is fully equipped with perishable and nonperishable items that patients require (in a nonmedical environment) and allows for scheduling flexibility that is not commonly available through in-center hemodialysis.²⁰⁷

Patient navigator programs have successfully improved screening, care access, and care processes for diverse patients in disease states such as in breast and colon cancer, diabetes, and chronic viral diseases.^{209,210} In fact, qualitative studies indicate that health system navigation is a high priority among diverse patients to assist with accessing care.^{145,211,212} Culturally trained peer navigators may be particularly effective in populations that experience cultural and/or linguistic barriers to care and those with limited health literacy.²¹³ For example, early findings from a culturally appropriate peer navigator intervention in Hispanic patients with KF suggest that such programs could enhance health-related quality of life.^{213,214} However, patient navigator interventions in CKD have yet to consistently demonstrate effectiveness across a range of kidney outcomes such as home dialysis utilization, transplant waitlisting, kidney transplantation, dietary adherence, and vascular access.^{215,216} More recent efforts that target dialysis facilities and aim to guide patients through the cumbersome medical work-up will likely require more time and continuous support to further reduce disparities in waitlisting or receipt of a kidney transplant among the intervention groups.²¹⁷⁻²²¹

Policy changes, particularly in kidney transplantation, have undoubtedly had the greatest impact in terms of population-level health equity. As gradient strategies, policy changes frequently target multilevel structural determinants of health inequities. For example, in 2013, amendments to the U.S. kidney allocation policy including the calculation of waiting time from the start of dialysis (or qualifying eGFR ≤ 20 mL/min) have led to improved rates of deceased-donor kidney transplantation, particularly for Hispanic and Black patients.^{168,186,222,223} The most recent 2021 data from the U.S. Scientific Registry of Transplant Recipients indicate that Black candidates now have the highest DDKT rate among racial groups.¹⁶⁸ In 2014, U.S. policymakers and health care providers lauded passage of the Affordable Care Act, which expanded basic health insurance coverage for low-income people in the United States (with variable, staggered adoption by individual states). In terms of kidney transplants, 59% more preemptive listings were observed among low-income people from states that expanded (vs. those that did not) basic health insurance coverage under the Act. There were also proportionately larger increases in basic health insurance coverage among preemptively listed people of minoritized race and ethnic groups as compared with those of non-Hispanic White race.^{139,224-226} From 2024, after years of lobbying, transplant recipients in the United States with federal health insurance will have their transplant immunosuppressive drugs covered indefinitely irrespective of age or disability status (previously transplant recipients in the United States lost their federal health insurance eligibility 36 months after the transplant if they did not

qualify for government disability status or were younger than 65).²²⁷ These efforts to address the structural determinants of kidney transplant disparities appear durable and will likely continue to decrease socioeconomic, geographic, and race and ethnic inequities in kidney transplantation over the long term.²²⁸

SUMMARY AND CHALLENGES

Socioeconomic, geographic, and race and ethnic disparities in the incidence and treatment of kidney diseases and their risk factors persist worldwide. Marked differences in education, living conditions, and employment opportunities combined with inequitable access to high-quality health care and genetic susceptibility act in tandem to promote and maintain disparate milieus for kidney disease. These conditions frequently intersect and lead to further reduction in kidney health among members of historically marginalized groups. Growing recognition of structural factors that underlie these disparities has fueled additional research and programs aimed at addressing barriers and unmet health-related needs of marginalized patients with kidney disease at multiple levels. However, despite in-depth knowledge of the root causes and consequences of these disparities, including their structural and social determinants, progress remains slow notwithstanding major disruption by the COVID-19 pandemic. In fact, widening inequalities and the “inverse” distribution of resources have moved the global nephrology community even further from health equity. In the setting of such widespread inequities, we should anticipate that disparities in kidney disease will continue to blight our community for the foreseeable future. Within nephrology, targeted efforts over the past 3 to 4 decades have yielded substantial declines in diabetes-related KF incidence among Indigenous people and improved access to deceased-donor kidneys among Black and Hispanic Americans. In the long term, reducing disparities rooted in social inequity will likely improve health and health care quality more than would technical advances in overall medical care.²²⁹ The social distribution of prematurity, low birth weight, diabetes mellitus, hypertension, and obesity and their indelible link to progressive declines in kidney function provides further evidence in favor of this approach to address disparities in nephrology.²⁷ Unfortunately, as laid bare by the COVID-19 pandemic, uncertainty surrounding the global economy and the rise of far-right populism continues to shroud potential legislative gains in health care for marginalized people.²²⁶ Needless to say, even in the presence of astute public policies aimed to address marked differences in social conditions, reducing long-standing disparities in nephrology will remain a major challenge for the foreseeable future.

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